

Alfredo Pasten Jr

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EXPERIENCE

- Teaching Assistant (Instructional Student Assistant)** | *CSULB – Python, SQLite* Aug. 2025 – Jan. 2026
- Designed and deployed a **SQLite**-backed tool to automate grade retrieval from **Zybooks** and synchronize records directly to Canvas, eliminating manual data entry for faculty.
 - Collaborated with faculty on course logistics and independently led proctoring responsibilities.

PROJECTS

- Multi-modal Agent Assistant Himiko** | *Ollama, ElevenLabs, PyTorch, Whisper, Discord* Ongoing
- Architected a modular, multi-modal AI agent that integrates with Discord, VTube Studio, and Minecraft, supporting real-time voice interaction via **Whisper** transcription and a custom **StyleTTS2** voice synthesizer.
 - Designed a plugin system enabling independent modules to be added without modifying core agent logic, demonstrating strong software architecture principles; e.g. **DeepL** translation for multilingual support.
- AI Agent Orchestrator** | *JS-Pybridge, Ollama, Docker* Oct. 2024 – Mar. 2025
- Developed a multi-agent AI framework enabling autonomous real-time **Minecraft** gameplay, inspired by and contributing to the open-source MindCraft project.
 - Containerized each agent using **Docker**, allowing multiple independent bots to run concurrently.
 - Bridged Python inference logic with JavaScript game APIs via **JS-Pybridge**
- ML Cryptocurrency Analyzer** | *PyTorch, Graph ML, Elliptic Dataset* May. 2026
- Led a graph-based machine learning project to classify illicit Bitcoin transactions using the Elliptic dataset, applying GNN architectures with **PyTorch**.
 - Achieved strong classification performance on an imbalanced financial graph, demonstrating applied expertise in ML for real-world compliance use cases.
- Server-side Minecraft Mod** | *Kotlin, Java, PaperAPI, Blockbench* Ongoing
- Engineered a full-scale **Minecraft** expansion – comprising a server plugin, datapack, and resource pack – that adds custom items, biomes, dimensions, and dungeons while remaining compatible with vanilla clients.
 - Published the plugin publicly and maintained an active server community of 40+ members; integrated the Dropbox API for seamless server-side asset delivery.

LEADERSHIP

- Association for Computing Machinery – Design Lead** Aug. 2023 – May. 2024
- California State University – Long Beach* Long Beach, CA
- Directed multiple design initiatives for the CS Department, including a centralized club information portal and a Discord authentication system for student and guest verification.
 - Produced promotional materials and event branding for general body meetings and large-scale events.
- Paul Merage Future Leaders Initiative** May. 2023 – Aug. 2023
- University of California – Irvine* Irvine, CA
- Selected for a competitive leadership program focused on emerging technologies; presented capstone work to the Mayor of Irvine, an NVIDIA Vice President, and leading industry executives.
 - Facilitated leadership workshops and built a cross-institutional network of peers and mentors.

EDUCATION

- California State University – Long Beach** Long Beach, CA
- Bachelor of Science in Computer Science, AI/ML Track* Aug. 2023 – Dec. 2026
- Hanyang University** Seoul, South Korea
- Hanyang International Winter School* Dec. 2025 – Jan. 2026
- Long Beach City College** Long Beach, CA
- Associate Studies in Computer Science* Aug. 2021 – Jun. 2023
- University of California – Berkeley** Berkeley, CA
- Undergraduate Studies in Electrical Engineering and Computer Science* Aug. 2020 – May. 2021

TECHNICAL SKILLS

- Languages:** Python, Kotlin, Java, TypeScript, JavaScript, C#, C++
- Frameworks:** Node.js, React, Next.js, Svelte, SvelteKit, Unity
- AI/ML:** PyTorch, Keras, Ollama, Whisper, HuggingFace, OpenAI API, Gemini API
- Developer Tools:** Git, Docker, Firebase, Vercel, Blockbench, Aseprite